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Surviving in semi-arid environments: functional coordination and trade-offs in shrubs from Argentina

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ABSTRACT

Human action has led to an increase in aridification, making water a progressively scarcer resource. In angiosperms, different species resolve this challenge in diverse ways, mainly through modifications of the xylem network, which is responsible for water efficiency and safety. Xerophytes generally show similar characteristics, but exceptions are rather frequent. One possible explanation for this lack of similarity among cohabiting species is that trade-offs and/or functional coordination between their organs occur and shape alternative survival strategies. Studying species that inhabit a common area can help to identify key traits that determine those diverse strategies and to predict which species might tolerate further environmental change.

We here examined the morpho-anatomical wood and bark traits of a group of species that live in a seasonally dry environment in Argentina. In a previous study, we described the leaf traits of these species and we thus aim to complement our findings and outline their strategies to manage water deficits. Our results show that there are different degrees of xeromorphism within this group. Clear xeromorphic traits, such as high vessel frequency and small diameter, were found in most species. However, some presented traits that were appropriate for mesic environments. An overview of leaf and wood traits indicates that the absence of a typical xeric characteristic in the wood might be compensated by the presence of a xeric leaf trait, and vice versa. Collectively, these trait combinations allow these species to survive in dry conditions and could influence their tolerance to increasing aridity.

Keywords: Xeric environment; morpho-anatomy; wood; bark; leaf.

INTRODUCTION

Water dictates the distribution of plants around the globe (Bradford & Hsiao 1982) and is the key element required for their most fundamental physiological processes (Boyer 1976). In angiosperms, the xylem network is specialized for transporting water throughout the body of the plant (Sperry 2003; Schweingruber & Börner 2018; Brodersen *et al.* 2019). According to the cohesion-tension theory, water movement through this network is driven by the negative pressure generated by evapotranspiration (Steudle 2001; Ángeles *et al.* 2004; Alemán-Sancheschúlz *et al.* 2019). However, conditions such as drought, where negative pressure is particularly high, increase the chances of xylem vessels becoming filled with air — a process called embolism (Sperry & Tyree 1988; Brodersen & McElrone 2013). Plants that survive in arid conditions have developed mechanisms to deal with this issue. Vessel diameter and length, for example, are in general negatively correlated with drought level, while the degree of vessel grouping, as well as vessel density, tend to increase with aridity (Fahn & Cutler 1992; Lindorf 1994; Carlquist 2009; Alemán-Sancheschúlz *et al.* 2019).

Xylem vessels might be the protagonist of resistance and tolerance to dry environments, but plants also depend on other characteristics pertaining to, for example, the leaves (Pittermann 2010) and the bark (Rosell 2019). Leaf traits such as location and density of stomata are some of the features that vary across environments and that often supplement modifications of the xylem vessels (Fahn & Cutler 1992; Lindorf 1994; Fu *et al.* 2019). Bark traits have received comparatively less attention

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in ecological studies of water deficit resistance, but evidence suggests that their role might be central in the hydraulic safety and efficiency trade-off (Loram-Lourenço *et al.* 2020). Generally associated with protection against fire and pathogens (Rosell 2016), the bark comprises, besides a dead outer layer, an inner layer composed of living cells. This layer shows the variation in thickness and density across humidity and temperature gradients, suggesting that storage of water and photosynthates, as well as protection of conductive elements, influence the ability to tolerate dry conditions (Rosell *et al.* 2014; Rosell 2019; Martín-Sanz *et al.* 2019; Loram-Lourenço *et al.* 2020).

The morpho-anatomical and physiological characteristics that we observe in plants today should be interpreted as snapshots of an integrative and ongoing process, whereby trait combinations, and most often trade-offs, optimize individual performance (Lachenbruch & McCulloh 2014). Traits considered beneficial are thought to increase the chances of survival to current environmental conditions, but also determine the ability to respond to future variations in the environment (Whitlock 2015), including those caused by climate change.

Human activities have had a major impact on natural ecosystems worldwide (Díaz *et al.* 2019). Drought, habitat fragmentation, compaction and humidity loss of soils, and increase of soil erosion are only a few of the negative consequences of land-use change. These detrimental effects are progressively turning forests and steppes into xeric and arid regions (Primack *et al.* 2001; Díaz *et al.* 2019). Due to the pervasiveness of this phenomenon (Vicente-Serrano *et al.* 2014; Diffenbaugh *et al.* 2015; Jiao *et al.* 2020), it is critical to explore the range of strategies that plants employ to survive in xeric conditions.

Environmental factors can have differential effects on phenotypic traits of diverse species (Kawecki & Ebert 2004) and, thus, plants are expected to resolve constraints, such as low water availability, in various ways (Miranda *et al.* 2010). Analysing which traits and trade-offs are common in xerophytes is crucial to infer which adaptations allow plants to manage the costs of aridification but also, and perhaps more importantly, to predict which species are more likely to tolerate further changes in their natural habitats (Fahn & Cutler 1992; Kawecki & Ebert 2004; Miranda *et al.* 2010; Lachenbruch & McCulloh 2014). Extreme drought affects gross primary production and ecosystem functioning (Zhang *et al.* 2019; Flach *et al.* 2020), particularly in plant communities dominated by less drought-tolerant species (Harrison *et al.* 2020). Therefore, characterizing a greater diversity of species and their strategies against drought is also instrumental for conservation biology.

Seasonally dry environments are ideal to explore strategies to survive in dry conditions as plants that live there are prone to suffer from embolism due to the high negative pressure. In this study, we examine five woody native species widely distributed in South America and that cohabit in a seasonally dry forest ecosystem in Central Argentina. This semi-arid area presents high levels of biodiversity (Cabido *et al.* 2018), particularly of deciduous and thorny trees and shrubs (Brown *et al.* 2006; Giorgis *et al.* 2011; Cabido *et al.* 2018) but has been subject to increasing levels of deforestation and degradation due to soybean cultivations and expansion of urban areas, with an approximate forest loss of 15% in the past 10 years (Global Forest Watch 2020).

In previous studies, the leaf morpho-anatomy of the same group of species was analysed (Delbón *et al.* 2007a, b, 2010). We there reported a series of structural, anatomical, and physiological characteristics that each species developed to survive in semi-arid environments. In general, the described traits were associated with the efficiency of water transport and avoidance of excessive water loss due to evapotranspiration (Fahn & Cutler 1992; Lindorf 1994). Within this framework, we aim to investigate the wood and bark morpho-anatomy of these five species and complement our findings with the leaf traits we previously described to characterize their global ecological anatomy. Specifically, we here try to address the following questions: (i) Which strategy related to water efficiency and safety is observed in the wood and bark of each species? (ii) Does the wood anatomy reflect a similar xeromorphic trend to the one observed in the leaves? (iii) Could potential trade-offs and functional coordination explain the prevalence of the patterns observed in the wood and the leaves of these species? (iv) Could these species tolerate higher levels of aridification and degradation compared to their natural habitats?

MATERIALS AND METHODS

Study area and species

We collected specimens of five native species from the 'Cerro El Cuadrado', Córdoba, Argentina (31°0.5′S, 64°27′W, 1100 m a.s.l.). The study area is a well-preserved relic of the Chaco Forest, where diverse native species coexist. This area has a semi-arid climate, characterized by seasonal thermal and precipitation amplitudes, daily thermal amplitude, and rocky soils (Brown *et al.* 2006; Giorgis *et al.* 2011; Cabido *et al.* 2018).

The species selected were *Baccharis aliena* (Spreng.) Joch.Müll (Asteraceae) (syn. *Heterothalamus alienus* (Spreng.) Kuntze), *Colletia spinosissima* J.F. Gmel. (Rhamnaceae), *Flourensia thurifera* (Molina) DC. (Asteraceae) (syn. *Flourensia campestris* Griseb.), *Kageneckia lanceolata* Ruiz & Pav. (Rosaceae) and *Schinus fasciculatus* (Griseb.) I.M. Johnst. (Anacardiaceae).

Baccharis aliena is a 1–2.5 m high shrub with many branches, thick and small leaves, distributed in central Argentina, Brazil, and Uruguay, and found up to 2500 m a.s.l. (Deble *et al.* 2005; Valtierra & Bonifacino 2014). *Colletia spinosissima* is a shrub of up to 4 m high, with branches terminated in thorns, and small ephemeral leaves, primarily distributed in Chile, Peru, Bolivia and Argentina at high elevations (approx. 4000 m a.s.l.) (Tortosa 1989, 2008; Zuloaga *et al.* 2008). *Flourensia thurifera* is a resinous shrub with well-developed leaves found in Argentina and Chile between 500 and 2500 m a.s.l. (Delbón *et al.* 2007a, b; Ospina *et al.* 2018). *Kageneckia lanceolata* is a shrub or tree of up to 5 m high, densely branched with deciduous leaves, distributed in Peru, Bolivia, Argentina, and Uruguay, but reaching up lowland valleys of 3500 m a.s.l. (Marticorena 2008; Romero 2019). *Schinus fasciculatus* is a shrub or tree up to 8 m high, with spiny branches and small leaves, distributed in Argentina, Brazil, and Paraguay between 0 and 1500 m a.s.l. (Barboza *et al.* 2006; Zuloaga *et al.* 2008).

At the beginning of the summer season (28 June 2008) five 1.5–2.5 m high, healthy-looking individuals of each species and separated by at least 5 m were randomly chosen, and branch samples were collected and fixed in FAA (10% Formaldehyde, Alcohol 50%, 5% Acetic acid and 35% water). We additionally collected herbarium specimens that were deposited in the Botanical Museum of Córdoba (CORD), Argentina; the collection numbers were Cosa 390–394 (for *C. spinosissima*, *B. aliena*, *K. lanceolata*, *S. fasciculatus* and *F. thurifera*).

Morpho-anatomical analysis of wood and bark traits

To standardize samples, we used sections that were 20–30 cm from the apex, from second and third-order branches, of 3–4 cm total diameter and with 2–3 growth rings, although only the last complete growth ring was used for observations. Distance to the branch top was standardized to compare the anatomical characteristics of different individuals and species (Anfodillo *et al.* 2013; Olson *et al.* 2018). Samples were cut using a sliding microtome in three different planes (transverse, tangential, and radial), stained with basic fuchsin — astra blue, and mounted in Canada balsam (Kraus *et al.* 1998; Zarlavsky 2014). For descriptions of the wood and bark, we used the terminology proposed by the IAWA Committee (1989) and Angyalossy *et al.* (2016). Images were taken using a Primo Star-Carl Zeiss light microscope and a Nikon Coolpix 5200 digital camera.

Morpho-anatomical analysis of leaf traits

The morpho-anatomical traits of the leaves that we included were taken from our previous studies (see Delbón *et al.* 2007a, b, 2010 for a complete description of the methodology). In brief, we collected leaf samples from the same individuals and prepared permanent microscope slides of transverse cuts of leaf blades and epidermal peels, following classical histological techniques (Zarlavsky 2014).

Analyzed traits and statistical elaboration

For the quantitative analysis of wood, six permanent transversal sections from five individuals per species were made. We counted vessels from the last complete growth ring and calculated vessel frequencies per sq. mm. Moreover, we prepared macerations of the wood and bark from each individual using Jeffrey's method described in Zarlavsky (2014) and took 10 measurements of the diameter and length of the vessels and libriform fibres. Statistical analyses were performed using R 4.0.0. We calculated the mean and standard deviations of each variable and performed one-way ANOVA and Tukey tests to compare each variable among species (Scheffé 1959; Sokal & Rohlf 1981). Vulnerability (IV) and mesomorphy (IM) indexes were determined using Carlquist's formulas (Carlquist 1977; Alemán-Sancheschúlz *et al.* 2019). IV = vessel element diameter/vessel frequency. IM = IV × length of vessel element.

To perform a quantitative analysis of the foliar epidermis we used data from our previous work (Delbón *et al.* 2007a, b, 2010). We analysed the following variables: stomatal frequency, epidermal cell frequency, and stomatal index for each leaf surface from the five same individuals from each species. Statistical analyses were done using R 4.0.0. We calculated the mean and standard deviations of each variable and performed one-way ANOVA and Tukey tests to compare each variable among species (Scheffé 1959; Sokal & Rohlf 1981).

A Principal Component Analysis (PCA) was carried out using the *prcomp* function in R using all the wood and leaf traits (Oskolski & Lowry 2001; Woods *et al.* 2005; Denham *et al.* 2006; Blanco-Dios 2007; MacLachlan & Gason 2010; Castello & Galetto 2013; Neyses & Hagman 2014). Twenty-five specimens (5 from each species) were analysed, each treated as a taxonomical operational unit (OTU). We used 11 continuous variables from wood and leaf: ECABF, abaxial epidermal cell frequency; ECADF, adaxial epidermal cell frequency; FL, fibre length; FW, fibre width; SABF, abaxial stomatal frequency; SADF, adaxial stomatal frequency; SABI, abaxial stomatal index; SADI, adaxial stomatal index; VL, vessel length; VF, vessel frequency; and VW, vessel width. Also, we used 12 categorical variables: AP, axial parenchyma; CT, cuticle thickness; FS, foliar structure; GR, growth ring boundaries; H, plant habit; LS, leaf size; P, porosity; PS, palisade/spongy ratio; R, rays; ST, stomatal type; TA, trichome abundance; and VD, vessel disposition, all previously coded as multistate (Tables 1 and 2). We used the *fviz_contrib*

Table 1.
Anatomical traits and quantitative analysis of wood.

	Baccharis aliena	*Colletia spinosissima*	*Flourensia thurifera*	*Kageneckia lanceolata*	*Schinus fasciculatus*
Growth ring	Less distinct	Intermediate	Distinct	Less distinct	Intermediate
Porosity	Diffuse-porous	Diffuse-porous	Ring-porous	Diffuse-porous	Diffuse-porous
Vessel disposition	Dendritic patterns	Solitary or in small groups	In clusters	Solitary	Diagonal patterns
Axial parenchyma	Abundant scanty paratracheal	Scarce scanty paratracheal	Abundant scanty paratracheal	Apotracheal axial	Scarce scanty paratracheal
Rays	1–2 seriate	2–3 seriate	3–5 seriate	1–2 seriate	2–3 seriate
Vessels/mm^2	236.8 ± 94.1 B	431.3 ± 93.9 C	140.5 ± 41.23 A	215.5 ± 51.6 B	631.4 ± 97.5 D
Vessel diameter (µm)	29.6 ± 8.51 AB	24.7 ± 7.4 A	42.8 ± 17.7 C	29.2 ± 6.3 AB	25.8 ± 7.4 A
Vessel length (µm)	178.2 ± 25.8 B	212.3 ± 40.4 C	156.2 ± 29.8 AB	473.1 ± 105.4 E	286.2 ± 72.3 D
Fibre diameter (µm)	14.5 ± 3.4 B	9.1 ± 2.3 A	15.2 ± 2.3 B	15.9 ± 3.1 B	14.5 ± 4.5 B
Fibre length (µm)	576.7 ± 47.4 C	511.2 ± 124.8 AB	565.3 ± 100.3 BC	736.9 ± 106.57 D	457.7 ± 119.8 A
VI	0.125	0.057	0.306	0.136	0.041
MI	22.28	12.1	47.8	64.33	11.74

Data are mean values ± standard deviation. Different letters indicate significant differences with Tukey tests, $p > 0.05$. VI, vulnerability index; MI, mesomorphy index.

function from the "factoextra" package in R to calculate variable contributions to the first two PCs. This function calculates the expected contribution of each variable if all had a uniform contribution. Variables that were above the cut-off line were retained for a second PCA analysis.

RESULTS

Morpho-anatomy of wood and bark

Growth ring boundaries were less distinct in *Baccharis aliena* (Fig. 1A), intermediate in *Colletia spinosissima* (Fig. 2A), distinct in *Flurensia thurifera* (Fig. 3A), less distinct in *Kageneckia lanceolata* (Fig. 4A), and intermediate in *Schinus fasciculatus* (Fig. 5A). All species have diffuse-porous wood, except for *F. thurifera* that exhibited ring-porous wood, as vessels in the earlywood were distinctly larger in diameter than those in the latewood (Fig. 3A). Vessels were exclusively solitary in *K. lanceolata* (Fig. 4A, B), solitary or in small groups of 2 or 3 in *C. spinosissima* (Fig. 2A, B), arranged in clusters in *F. thurifera* (Fig. 3A), in diagonal patterns in *S. fasciculatus* (Fig. 5A) and in dendritic to polygonal patterns in *B. aliena* (Fig. 1A, B). Vessel frequency was very high in *C. spinosissima* and *S. fasciculatus*, low in *F. thurifera*, and intermediate in the remaining species (Table 1).

Vessels were generally observed in association with tracheids and axial parenchyma. In all species scanty paratracheal axial parenchyma was observed, being more abundant in *F. thurifera* (Fig. 3B) and *B. aliena* (Fig. 1B). In *K. lanceolata* apotracheal axial parenchyma was diffuse with a few single parenchyma cells irregularly distributed among the fibrous elements (Fig. 4B). Abundant libriform fibers were observed in all species, but septate fibers were present only in *S. fasciculatus* (Fig. 5B).

Rays were 1–2 seriate in *B. aliena* (Fig. 1B, C) and *K. lanceolata* (Fig. 4B, C), 2–3 seriate in *C. spinosissima* (Fig. 2B, C) and *S. fasciculatus* (Fig. 5B, C), while 3–5 seriate rays were common in *F. thurifera* (Fig. 3B, C). Ray height varied between 10 and 15 cells. Rays were heterogeneous with procumbent body ray cells and one row of upright marginal cells in *C. spinosissima* (Fig. 2D), *K. lanceolata* (Fig. 4D) and *S. fasciculatus* (Fig. 5D), whereas in *F. thurifera* (Fig. 3D) and *B. aliena* (Fig. 1D) rays were homogeneous, with only procumbent body ray cells. Tannins, druses, and prismatic crystals were present in *S. fasciculatus* rays (Fig. 5B–D, F), while tannins were presumably found in *K. lanceolata* (Fig. 4B–D).

We found pith flecks only in *C. spinosissima*, which are anomalies of the parenchymal cells generally associated with growth rings (Fig. 2A). In longitudinal sections, we observed columns of parenchymal tissue (pith flecks) arranged along the longitudinal axis of the plant (Fig. 2E).

All species showed parenchymatic piths, with crystals and presumably tannins present only in *S. fasciculatus* (Fig. 5G). *F. thurifera* had numerous resinous secretory ducts (Fig. 3E). Furthermore, vessels of the first growth ring and primary xylem were infiltrated with a resinous substance similar to that secreted by ducts (Fig. 3E). Tyloses in vessels were observed in *S. fasciculatus* (Fig. 5E).

Table 2.
Habit, morpho–anatomical traits and quantitative analysis of leaf.

	Baccharis aliena	*Colletia spinosissima*	*Flourensia thurifera*	*Kageneckia lanceolata*	*Schinus fasciculatus*
Plant habit	Evergreen shrub	Thorny bush with ephemeral leaves	Semi-deciduous shrub	Semi-deciduous shrub or tree	Evergreen shrub or tree
Leaf shape	Linear	Elliptic–ovate	Aovate–lanceolate	Oblong–lanceolate	Spatulate
Leaf size (cm) (length/width)	0.5–1.7/0.3–0.5	0.6–1/0.5–0.7	3–4/2–3	3–5/1–1.5	1–1.5/0.8–1
Stomatal type	Paracytic	Paracytic	Anomocytic	Anomocytic	Anomocytic
Trichome type/abundance	Glandular abundant	Non-glandular scarce	Non-glandular and glandular abundant	Without trichomes	Non-glandular and glandular abundant
Cuticle thickness	Similar in both epidermis	Thicker in adaxial	Similar in both epidermis	Similar in both epidermis	Thicker in adaxial
Foliar structure	Isobilateral	Dorsiventral	Isobilateral	Dorsiventral	Dorsiventral
Palisade parenchyma	3–5 strata compact	2–4 strata medium compact	2–3 strata little compact	2 strata compact	2–3 strata compact
Spongy parenchyma	1 strata compact	6–8 strata lax	1–2 strata lax	4–5 strata lax	4–5 strata compact
Palisade/spongy	3/1	1/3	4/1	1/1	1/1
Bundle sheath and extensions	Both developed	Not visible	Both developed	Only sheath developed	Not visible
Other sharacteristics	Schizogenic resin ducts and tannins	Druses	Schizogenic resin ducts	–	Lysogenic resin ducts, druses and tannins
Adaxial stomatal frequency	71.5 ± 30.7 A	–	170.8 ± 27.2 B	–	–
Abaxial stomatal frequency	63.9 ± 35 A	173.1 ± 74.5 BC	198 ± 35.4 CD	232.2 ± 63 D	133.9 ± 38.6 B
Adaxial epidermal cell frequency	1567.3 ± 463 A	–	1343.8 ± 136.3 A	–	–
Abaxial epidermal cell frequency	1637.8 ± 434.1 C	2711.5 ± 367.4 D	1229.2 ± 125.7 B	1310.4 ± 245.9 B	409.5 ± 40.5 A
Adaxial stomatal index	4.4 ± 0.4 A	–	11.3 ± 0.7 B	–	–
Abaxial stomatal index	3.8 ± 0.6 A	6.1 ± 0.8 A	13.9 ± 1.1 B	15.1 ± 0.6 B	24.7 ± 0.7 C

Extracted from Delbón *et al.* (2007a, b, 2010). *Baccharis aliena* = *Heterothalamus aliena* and *Flourensia thurifera* = *Flourensia campestris*. Data are mean values ± standard deviation. Different letters indicate significant differences with Tukey tests, $p > 0.05$.

Developed periderm was present in the bark of all species, consisting of a few layers of phellem and a single layer of phelloderm, internal to the phellogen (Figs 3F, 4E, 5H), with the exception of *C. spinosissima*. This species retained the epidermis or developed a few layers of periderm with abundant photosynthetic chlorenchyma in the inner bark (Fig. 2F). All of our species presented a developed inner bark with living cells and abundant fibre bundles dispersed in the inner bark and around the secondary phloem. Secretory ducts were found in the bark of *F. thurifera* (Fig. 3F) and *S. fasciculatus* (Fig. 5H), while they were observed as part of the phloem rays in *K. lanceolata* (Fig. 4E, F). *S. fasciculatus* presented druses, crystals, and possibly tannins (Fig. 5H).

Baccharis aliena presented a different secondary phloem anatomy relative to the other species (Fig. 1E, F). In this species, the conducting elements — sieve tube members with companion cells — were arranged in small groups that were completely surrounded by abundant libriform fibres forming a protective sheath. Parenchyma cells and phloem rays were found among the fibre bundles (Fig. 1F).

Vessel elements with helical thickening, bordered pits, and alternate and simple perforation plates were observed in macerations (Fig. 4G). Libriform fibres with simple pits were found in all species, being septate in *S. fasciculatus* (Fig. 5F). Imperforate tracheids with bordered pits, particularly abundant in *K. lanceolata*, were also observed (Fig. 4G).

Morpho-anatomy of the leaf

Morpho-anatomical differences in leaf traits were reported in our previous study (Delbón *et al.* 2007a, b, 2010) (summary of key traits; Table 2 and Fig. 6).

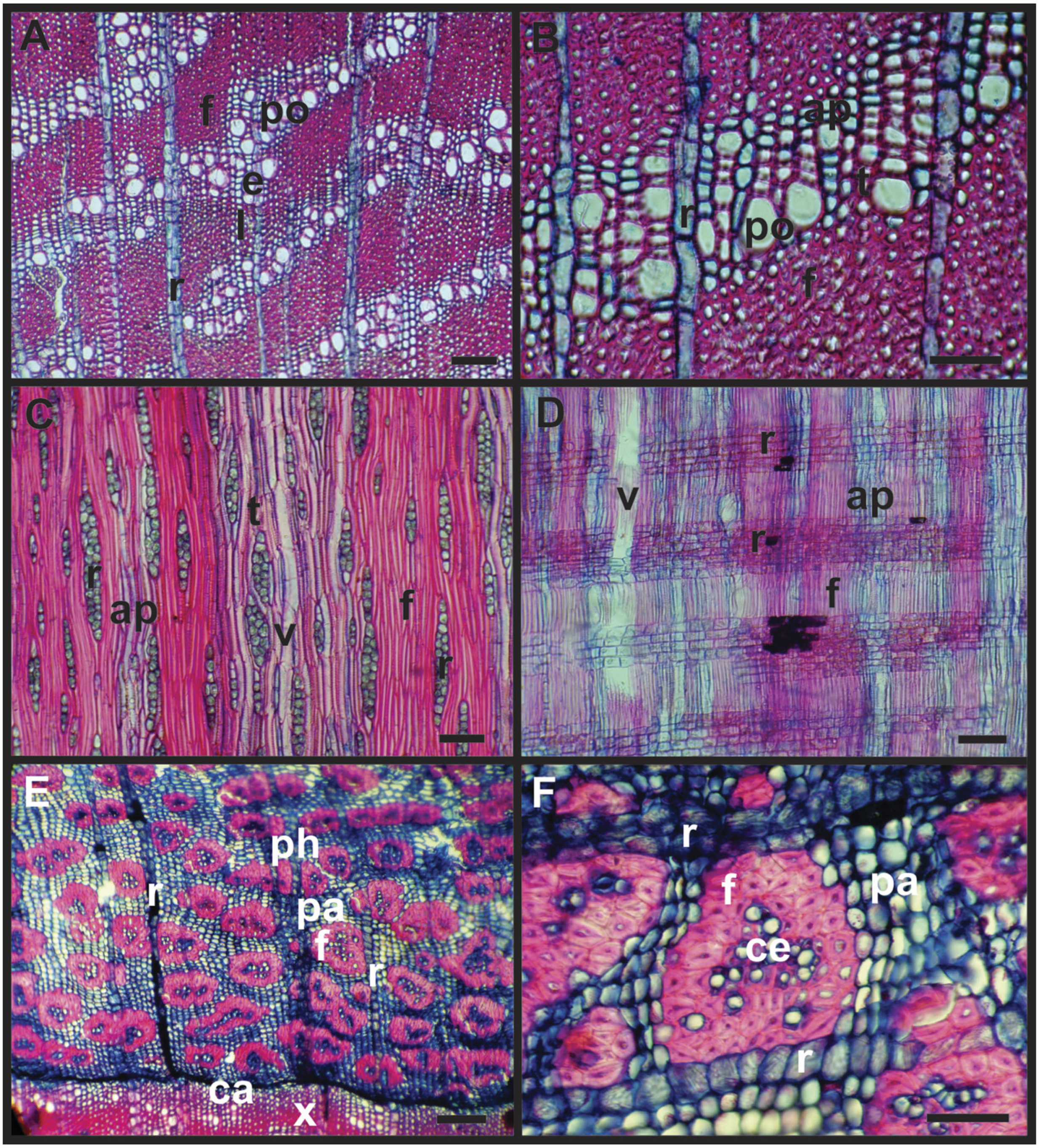

Figure 1. Wood and bark in *Baccharis aliena*. (A) wood in transversal section showing diffuse porosity. (B) detail of vessels in dendritic pattern. (C) tangential section showing 1–2 seriate rays. (D) radial section showing homogeneous rays. (E) bark in transversal section. (F) detail of conductive elements surrounded by fibres. Abbreviations: ap, axial parenchyma; ca, cambium; ce, conductive elements; e, earlywood; f, fibre; l, latewood; pa, parenchyma; ph, phloem; po, pore; r, ray; t, tracheids; v, vessel; x, xylem. Scale bars: B, F, 50 µm; A, C–E, 100 µm.

In summary, we observed different degrees of numerous adaptive leaf traits. In *B. aliena* the small leaf area (Table 2) and thick palisade parenchyma indicate a high degree of xeromorphism (Fig. 6B, C); in *C. spinosissima* the xeromorphic characteristics are less marked (Fig. 6E, F), but the leaves are quickly deciduous, and assimilation is practically restricted to the stem (Table 2). *S. fasciculatus* follows in the adaptive trend due to its small leaves (Table 2) and some xeromorphic anatomical traits (Fig. 6N, O). Finally, *F. thurifera* and *K. lanceolata* with their larger leaves (Table 2) exhibit anatomical structures (Fig. 6H, I, K, L) showing minimal xeromorphic traits.

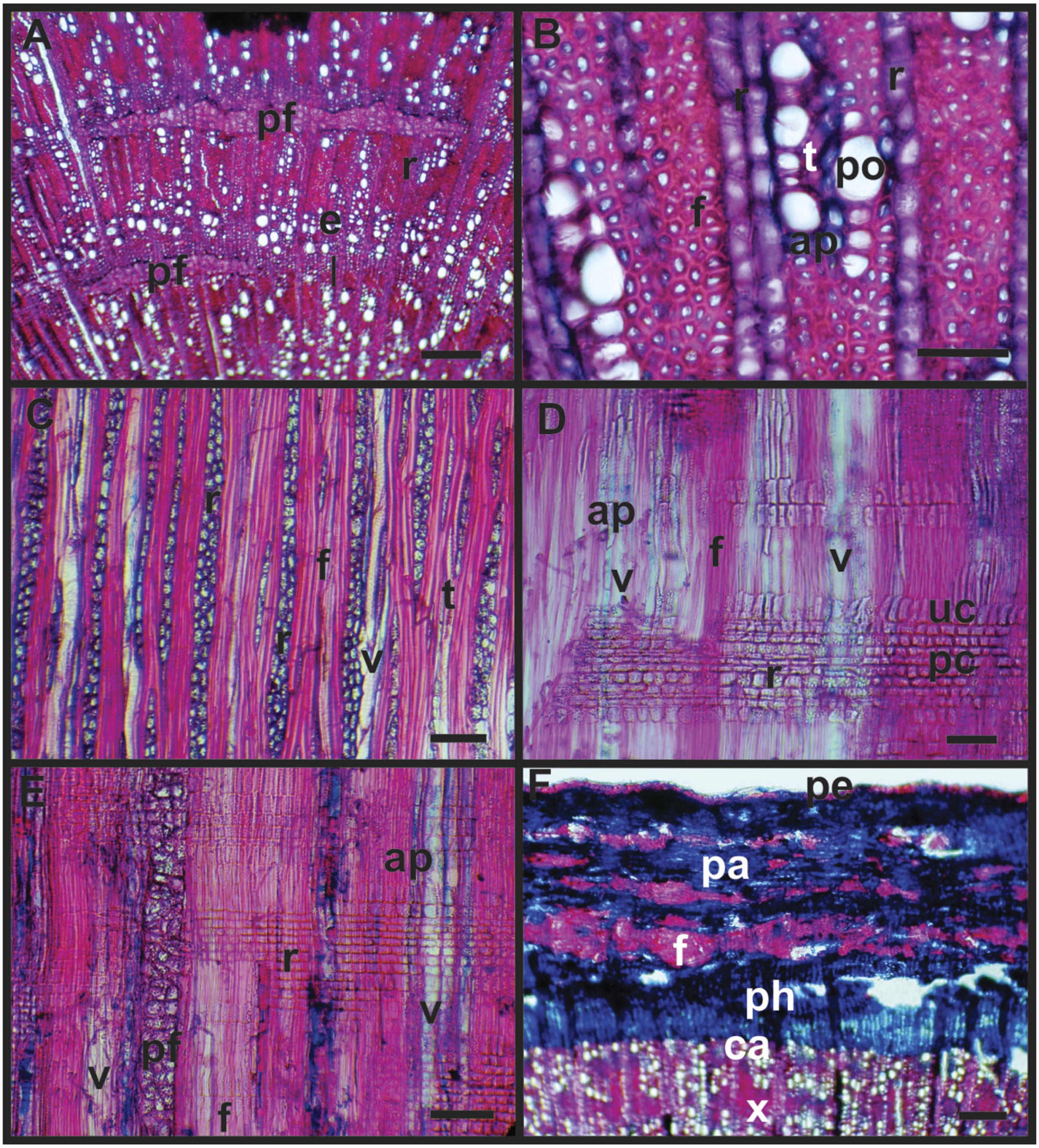

Figure 2. Wood and bark in *Colletia spinosissima*. (A) wood in transversal section showing diffuse porous and pith flecks. (B) detail of vessels group. (C) tangential section showing 2–3 seriate rays. (D) radial section showing a heterogeneous ray. (E) longitudinal section of pith flecks. (F) bark in transversal section with undeveloped periderm. Abbreviations: ap, axial parenchyma; ca, cambium; ce, conductive elements; e, earlywood; f, fibre; l, latewood; pa, parenchyma; pc, procumbent cell; pe, periderm; pf, pith fleck; ph, phloem; po, pore; r, ray; t, tracheids; uc, upright cell; v, vessel; x, xylem. Scale bars: B, 50 μm; C–E, 100 μm; A, F, 150 μm.

Quantitative traits

The statistical analysis of wood traits showed significant differences for all variables (Table 1 and Fig. 7). *F. thurifera* presented the shortest vessels, while *K. lanceolata* and *S. fasciculatus* had the longest, but also a greater size variability, reflected by their larger interquartile range and greater maximum values (Fig. 7A). *C. spinosissima* and *S. fasciculatus* had smaller vessel diameter (Fig. 7B), IV and IM indexes, and also greater vessel frequency (Table 1), all of which correspond to xeromorphic plants (Table 3).

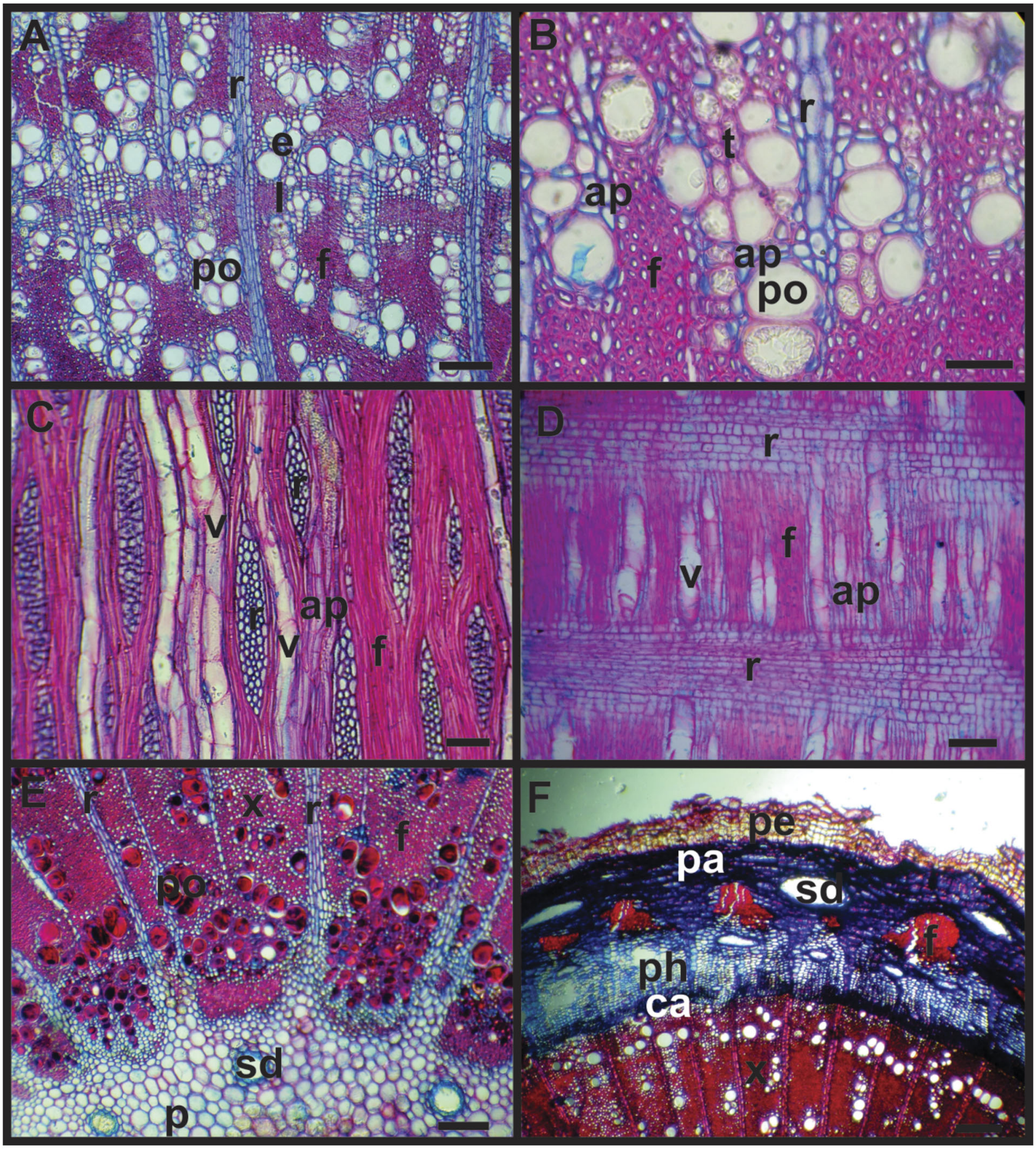

Figure 3. Wood and bark in *Flourensia thurifera*. (A) Wood in transversal section showing ring porous. (B) detail of vessel clusters. (C) tangential section showing 3–5 seriate rays. (D) radial section showing homogeneous rays. (E) pith with secretory ducts and obliterate xylem vessels. (F) bark in transversal section with secretory ducts and periderm developed. Abbreviations: ap, axial parenchyma; ca, cambium; e, earlywood; f, fibre; l, latewood; p, pith; pa, parenchyma; pe, periderm; ph, phloem; po, pore; r, ray; sd, secretory ducts; t, tracheids; v, vessel; x, xylem. Scale bars: B, 50 µm; A, C–E, 100 µm; F, 150 µm.

Conversely, *F. thurifera* presented a significantly larger vessel diameter, but also a large standard deviation (Table 1), larger interquartile range, and the highest overall value (Fig. 7B); this is directly related to its ring porosity, characterized by large early-wood and small late-wood vessels. It also presented smaller vessel frequency, and the highest IV index (Table 1).

Schinus fasciculatus and *K. lanceolata* presented the shortest and longest fibres, respectively, although they were septate libriform in the former and fibre tracheids in the latter (Fig. 7C); in *K. lanceolata* the fibre tracheids are only slightly longer than the vessels (Fig. 7A, C). The diameter of the fibres was similar among species, except in *C. spinosissima* (Fig. 7D).

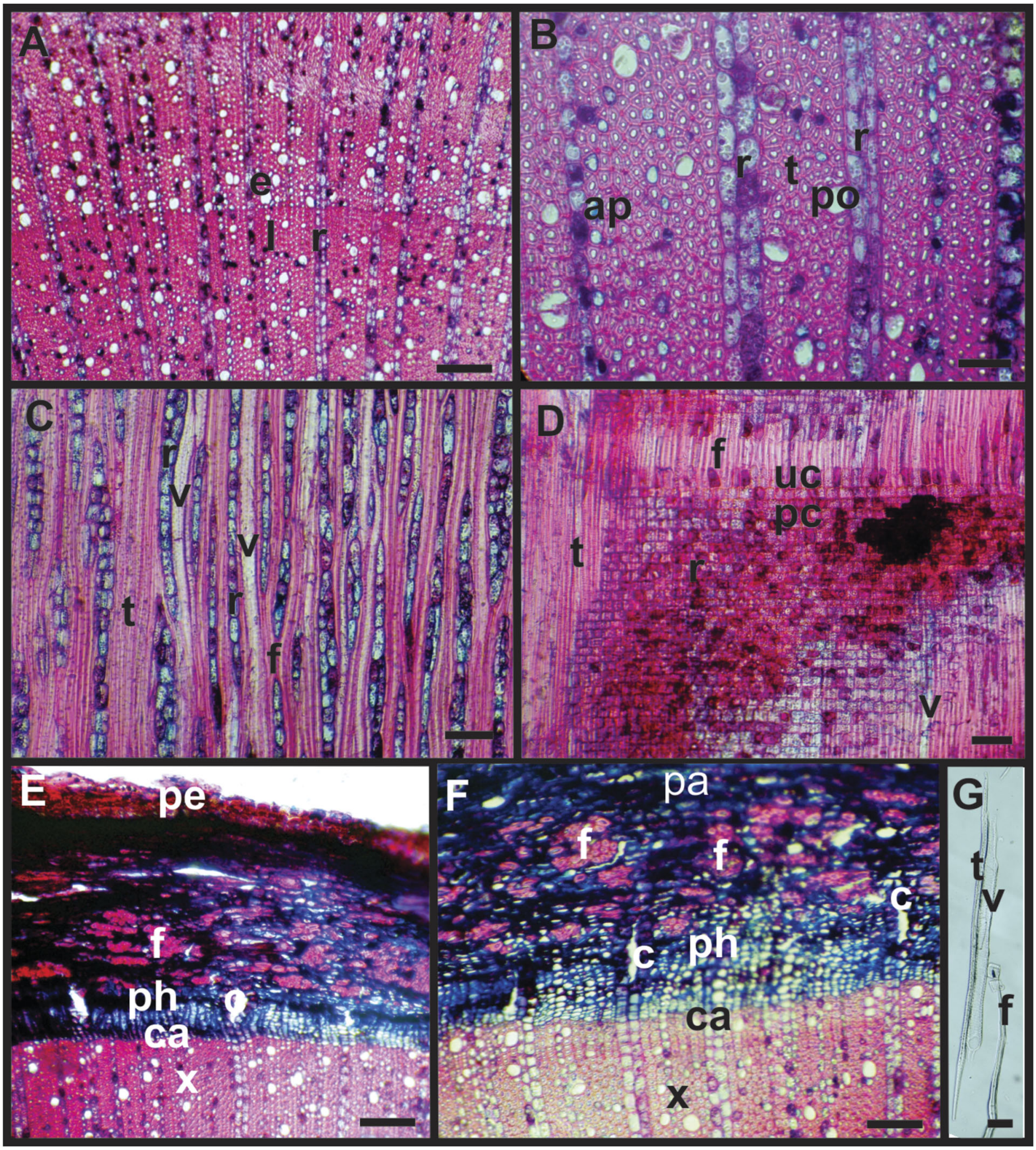

Figure 4. Wood and bark in *Kageneckia lanceolata*. (A) wood in transversal section showing diffuse porous. (B) detail of solitary vessels and diffuse apotracheal axial parenchyma. (C) tangential section showing 1–2 seriate rays. (D) radial section showing heterogeneous rays. (E) bark in transversal section. (F) detail of phloem radial canals. (G) tracheid, vessel, and fibre from maceration. Abbreviations: ap, axial parenchyma; c, canal; ca, cambium; e, earlywood; f, fibre; l, latewood; pa, parenchyma; ph, phloem; po, pore; r, ray; t, tracheids; v, vessel; x, xylem. Scale bars: B, G, 50 µm; C, D, F, 100 µm; A, E, 150 µm.

Regarding the leaves (Fig. 8 and Table 2), the most relevant quantitative trait is the absence of stomata on the adaxial surface of *C. spinosissima*, *K. lanceolata* and *S. fasciculatus* (Fig. 8A), the latter also presenting less stomatal frequency on the abaxial surface (Fig. 8B). *Baccharis aliena* and *F. thurifera* present stomata on both leaf surfaces (Fig. 8A, B), but *B. aliena* presented a very low stomatal frequency, while *F. thurifera* had a high frequency of both.

In all species, the values of the stomatal indexes are consistent with the results of the stomatal frequencies (Fig. 8C, D), except in *S. fasciculatus*. In this species, the stomatal abaxial index is visibly higher than the other species (Fig. 8D), as a

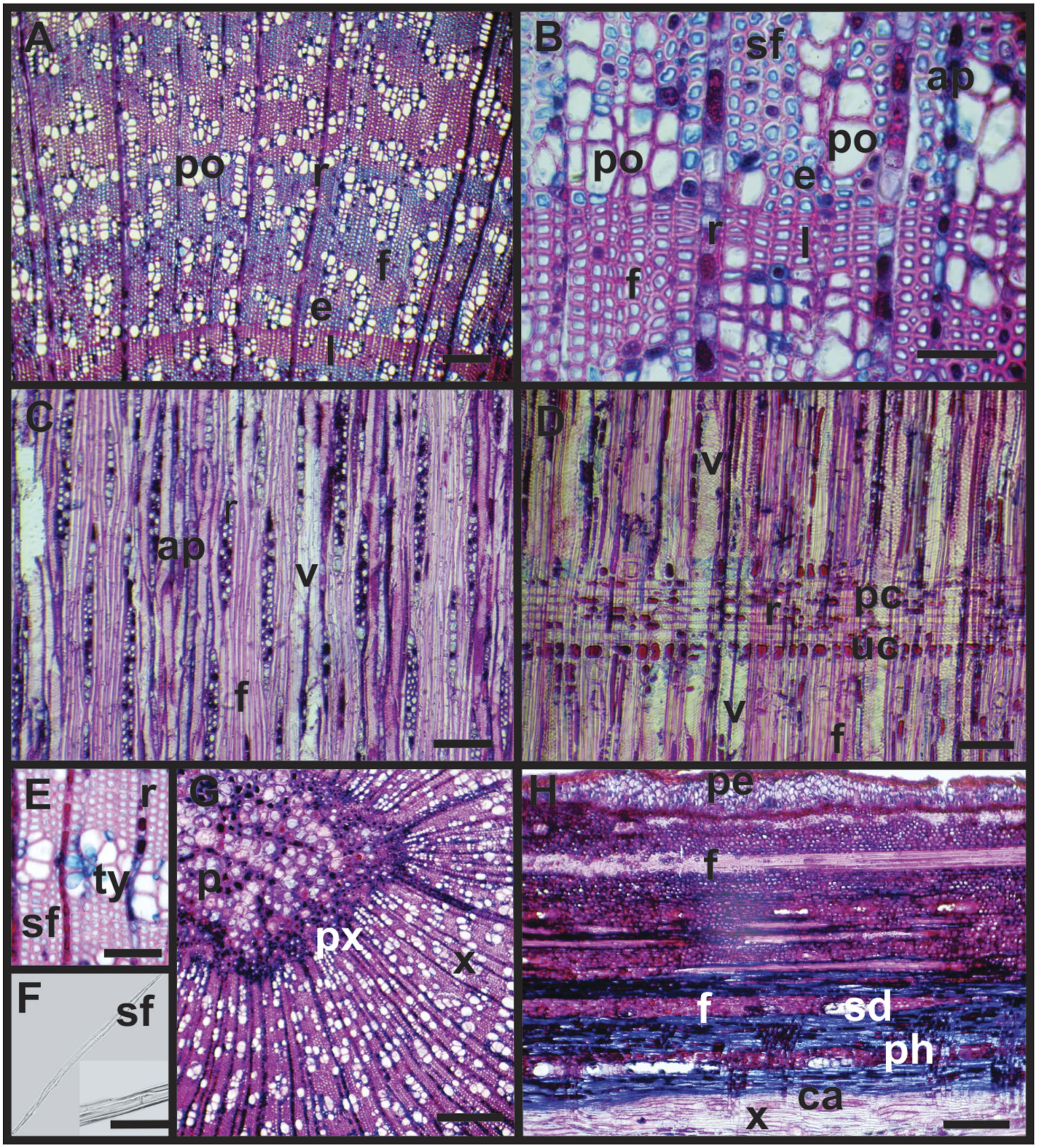

Figure 5. Wood and bark in *Schinus fasciculatus*. (A) wood in transversal section showing diffuse porous. (B) detail of vessel clusters. (C) tangential section showing 2–3 seriate rays. (D) radial section showing heterogeneous rays. (E) detail of vessels with tyloses. (F) septate fibre from maceration. (G) pith with crystals and tannins. (H) bark in longitudinal section with secretory ducts and periderm developed. Abbreviations: ap, axial parenchyma; ca, cambium; e, earlywood; f, fibre; l, latewood; p, pith; pa, parenchyma; pc, procumbent cell; pe, periderm; ph, phloem; po, pore; px, primary xylem; r, ray; sd, secretory ducts; sf, septate fibres; t, tracheids; ty, tyloses; uc, upright cell; v, vessel; x, xylem. Scale bars: B, E–F, 50 µm; A, C, D, 100 µm; G, H, 150 µm.

consequence of the very low frequency of epidermal cells and their large size (Table 2). *Colletia spinosissima* presents more xeromorphic characters in the leaves (Table 3).

In the PCA, the first two components explained 62.8 % of the variation (37.7 and 25.1%, respectively) (results not shown). Analysing the contribution of each variable to the first two components, 9 wood and leaf traits for PC1 and 11 for PC2 had contributions above the cut-off line (Figs A1 and A2 in the Appendix). A second PCA using these 20 traits only, showed

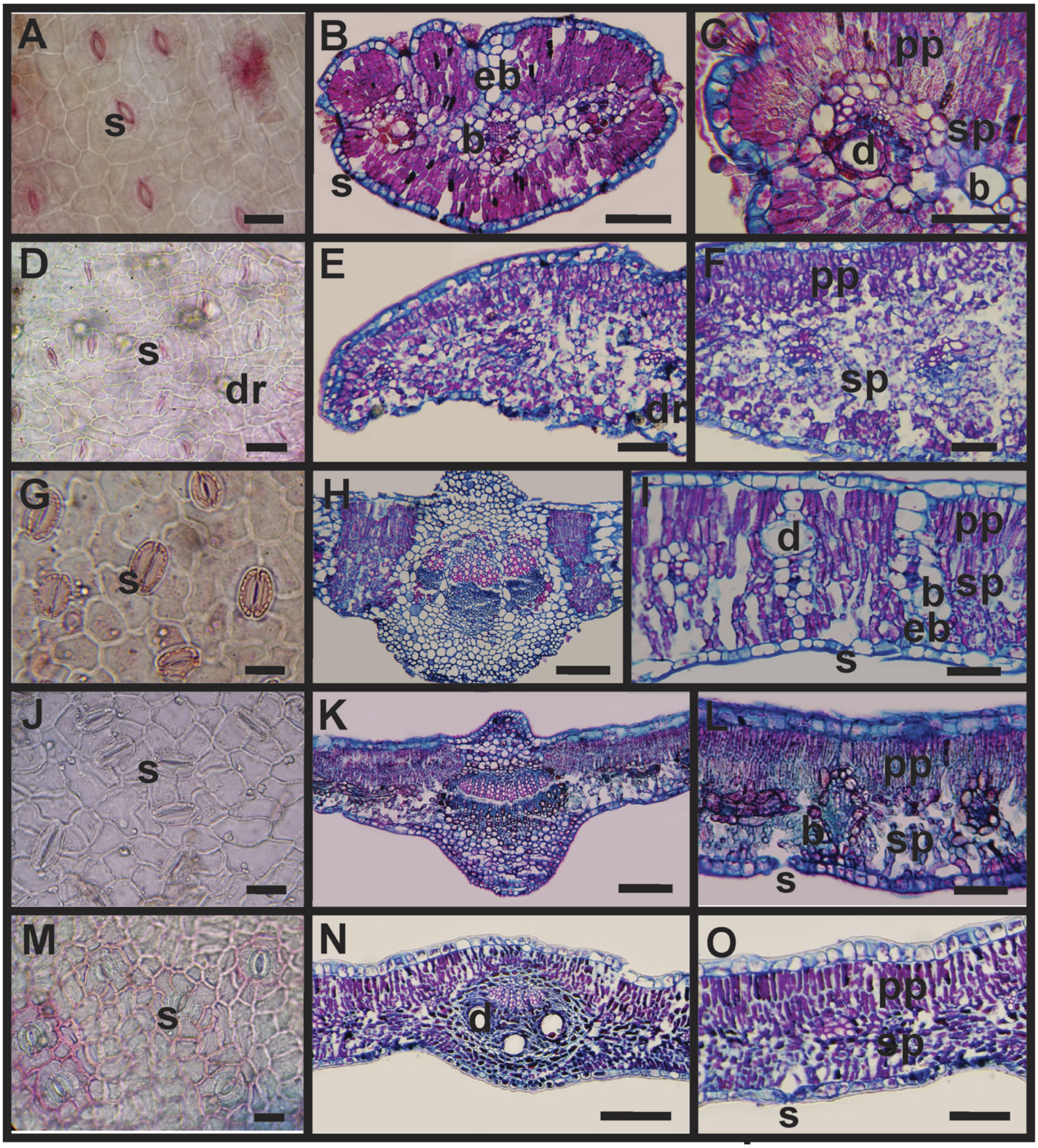

Figure 6. Leaf in (A, B) *Baccharis aliena*, (D–F) *Colletia spinosissima*, (G–I) *Flourensia thurifera*, (J–L) *Kageneckia lanceolata*, (M–O) *Schinus fasciculatus*. (A, D, G, J, M) Surface view of the epidermis; (B, E, H, K, N) cross-section of leaf lamina and midribs; (C, F, I, L, O) detail of mesophyll. Abbreviations: b, bundle sheath; d, duct; dr, druse; eb, extensions of bundle sheath; pp, palisade parenchyma; s, stomata; sp, spongy parenchyma; t, trichome. Scale bar: A, D, G, J, M, 30 µm; B, E, H, K, N, 150 µm; C, F, I, L, O, 80 µm.

that the first two PCs explained 74.0% (42.3 and 31.7%, respectively) of variation (Fig. 9). All of the species were spatially distant, except for *C. spinosissima* and *S. fasciculatus*. The second dimension clearly separated *K. lanceolata* and *F. thurifera* from the rest, reflecting a xeromorphic gradient that runs from species with markedly xeromorphic traits (*C. spinosissima*, *S. fasciculatus* and *B. aliena*) to those with intermediate trait combinations (*F. thurifera*), to the least xeromorphic species (*K. lanceolata*). The variables that heavily contribute to PC axis 2 and explain this xeromorphic gradient are a mixture of wood and leaf traits: fibre length (FL) and width (FW), vessel length (VL), abaxial stomatal frequency (SABF), leaf size (LS), axial parenchyma (AP), stomatal type (ST), cuticle thickness (CT), vessel disposition (VD) and trichome abundance (TA) (Figs 9

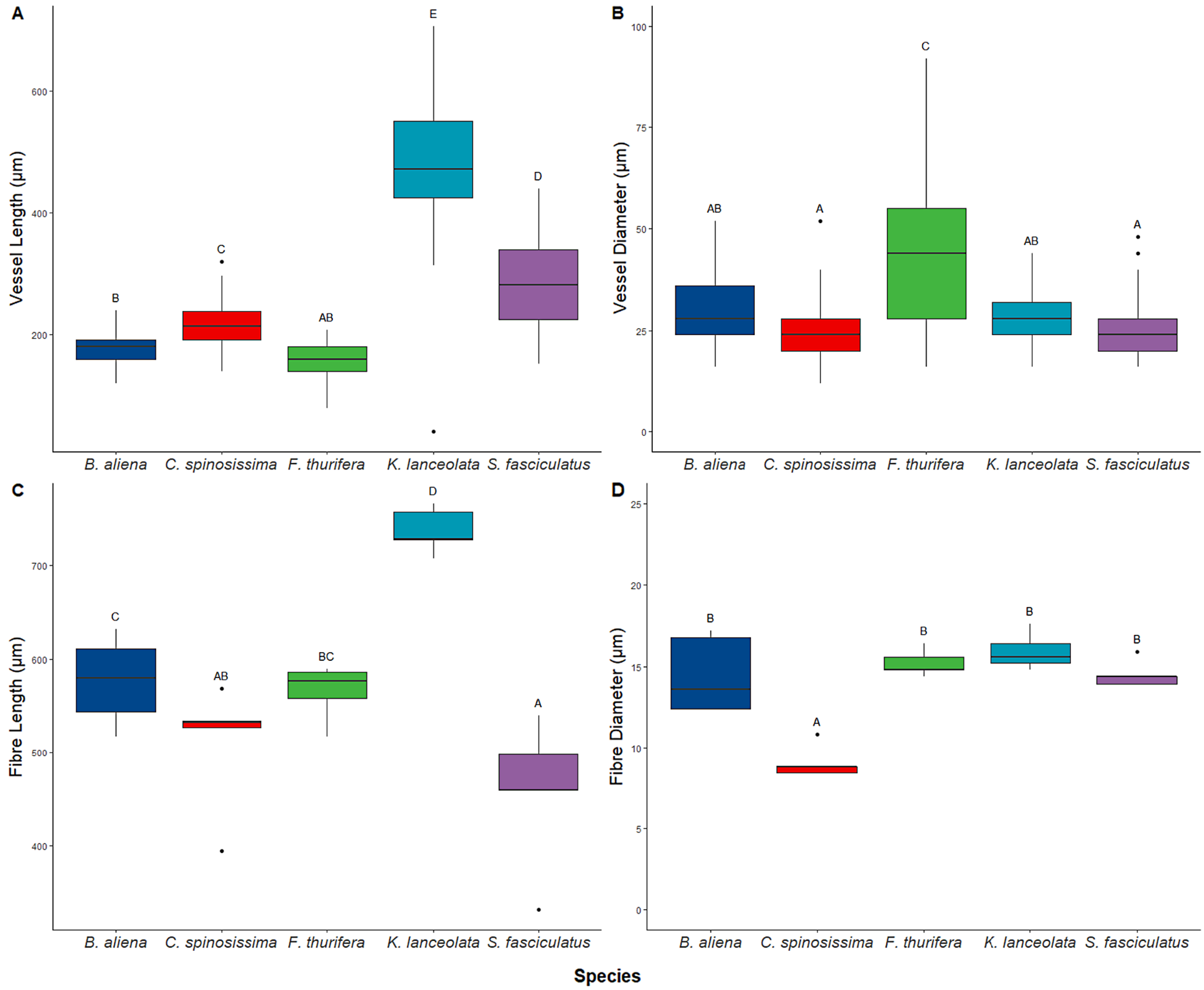

Figure 7. Quantitative wood analysis. Boxplots of trait values for the five species examined. Significance of each ANOVA is shown under the corresponding boxplot. Different letters indicate significant differences with Tukey tests, $p < 0.05$.

and A2). Moreover, on the first PC, *B. aliena* and *F. thurifera* were closer to each other and this could be explained by the presence of adaxial stomata on the leaf, i.e. adaxial stomatal frequency and index, and adaxial epidermal cell frequency (Figs 8A, C and A1).

DISCUSSION

Environmental constraints pose challenges to organisms that must be overcome in order to survive. Strategies to minimize the costs of unsuitable conditions are not universal and studying the diverse ways in which plants manage harsh conditions, such as those resulting from drought and aridification, is necessary to identify the key traits shared by members of different species and those that are distinct and unique to a handful of them (Bucci *et al.* 2004; Lachenbruch & McCulloh 2014). However, since modifications that we observe in plants are part of an integrative process, a single plant trait should not be studied in isolation, but rather in conjunction with other traits to recognize possible trade-offs that may not be evident otherwise (Sobrado 1993; Miranda *et al.* 2010; Díaz *et al.* 2016; Fu *et al.* 2019). The species studied here showed morpho-anatomical differences in their wood and bark that reflect different degrees of xeromorphism. When combined with the leaf traits we described in previous studies, this group of plants exemplifies various strategies that, given their widespread distribution, seem to be effective in semi-arid conditions.

Xylem traits reflect underlying trade-offs related to efficiency and safety of water transport (Lindorf 1994; Hargrave *et al.* 1995; Baas 2004; Carlquist 2009; Alemán-Sancheschúlz *et al.* 2019). In this regard and following the classification proposed by Lindorf (1994), our species presented extremely numerous vessels (>150 vessels/mm²) and an extremely small vessel diameter (<30 µm), except for *F. thurifera* that presented very numerous vessels (40–150 vessels/mm²) and a very small vessel diameter

Table 3.
Summary of key leaf and wood traits in relation to their associated degree of xeromorphism.

	Leaf			Wood					
	Size/shape	Stomatal traits[1]	Other traits[2]	Vessel			Fibre traits[3]	VI	MI
				Frequency	Diameter	Length			
Baccharis aliena	X	O	X	X	X	X	X	X	X
Colletia spinosissima	X	X	X	X	X	X	X	X	X
Flourensia thurifera	O	X	X	O	O	X	X	X	X
Kageneckia lanceolata	O	X	O	X	X	O	O	X	X
Schinus fasciculatus	X	O	X	X	X	X	X	X	X

Except for *C. spinossissima* that shows all xeromorphic traits, the rest of the species present at least one trait that is not generally observed in plants that live in arid environments. Trade-offs and/or functional coordination could be underlying the water efficiency and safety strategies used in these species. X, trait generally associated with the xeric environment; O, trait generally associated with the mesic environment.

[1] Stomatal frequency, location and index.

[2] Presence of blade extensions, sheath, trichomes, tannins or druses.

[3] Fibre length and diameter.

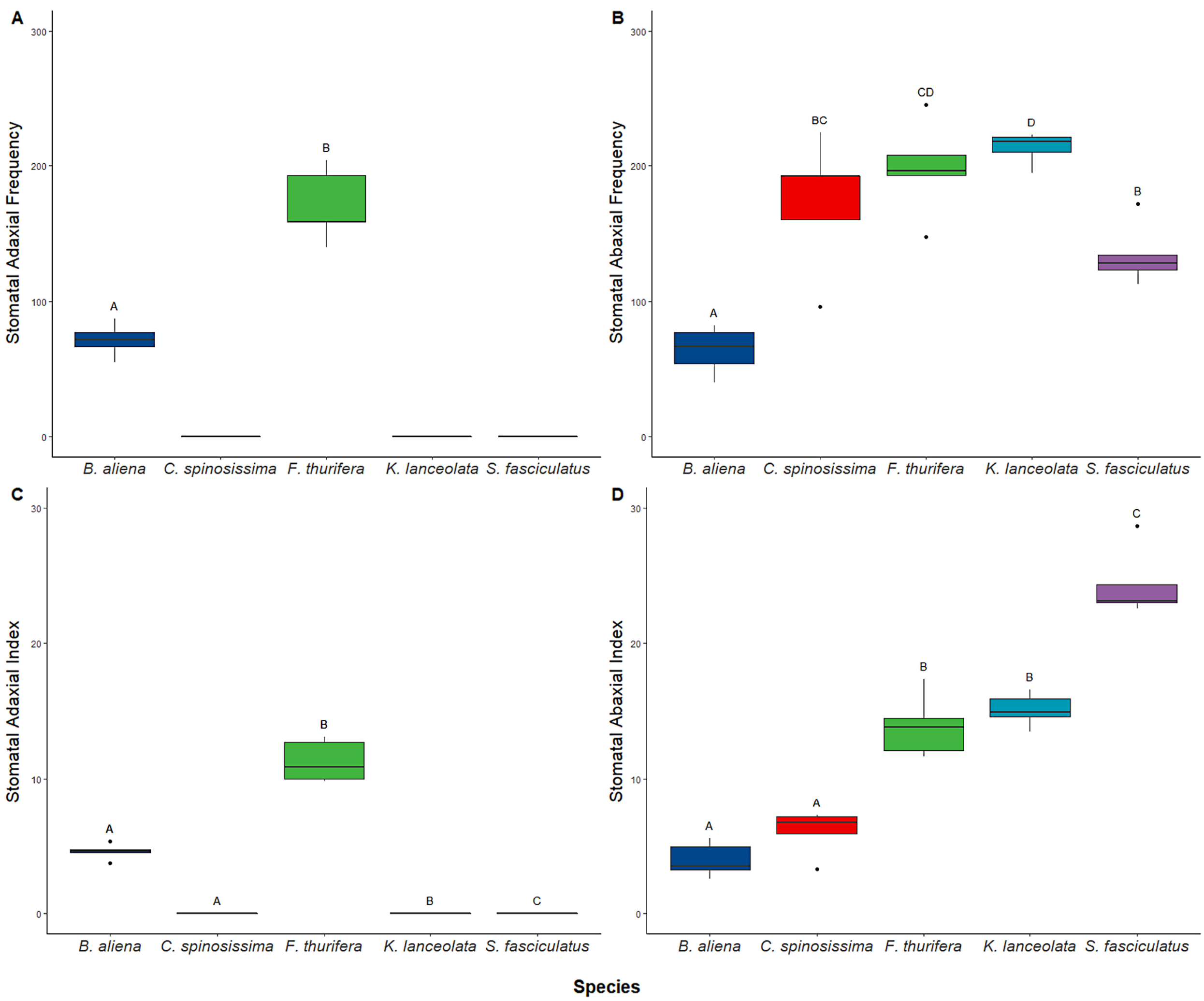

Figure 8. Quantitative leaf analysis. Boxplots of trait values for the five species examined. Significance of each ANOVA is shown under the corresponding boxplot. Different letters indicate significant differences with Tukey tests, $p < 0.05$.

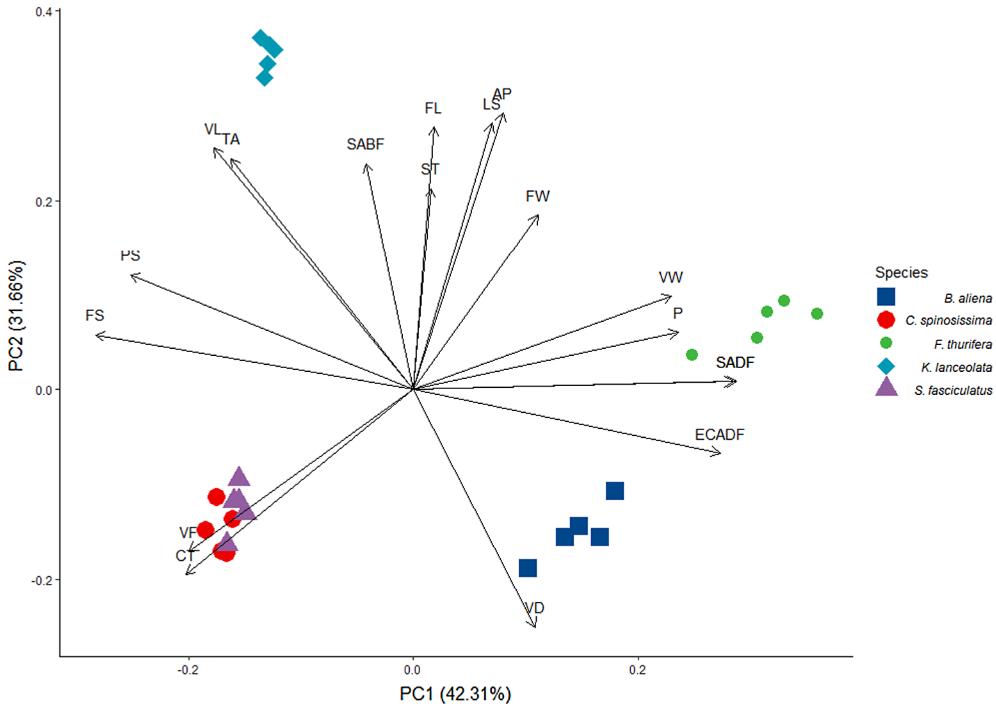

Figure 9. PCA for the five species examined. Plot of all specimens (25 OTUs) and leaning of the most influential characters represented on the first two principal components resulting from principal component analysis. Abbreviations of variables: AP, axial parenchyma; CT, cuticle thickness; ECADF, adaxial epidermal cells frequency; FL, fibre length; FS, foliar structure; FW, fibre width; LS, leaf size; P, porosity; PS, palisade/spongy ratio; SABF, abaxial stomatal frequency; SADF, adaxial stomatal frequency; ST, stomatal type; TA, trichome abundance; VD, vessel disposition; VF, vessels/mm²; VL, vessel length; VW, vessel width.

(30–50 µm). These results are not common and do not follow the general pattern found in species that inhabit similar regions. Several studies on the tree species *Prosopis*, *Lithraea*, *Schinopsis*, and *Aspidosperma* from arid regions in North and Central Argentina reported higher vessel diameter values and lower vessel frequencies (Martijena 1987; Villagra & Roig Juñent 1997; Bolzón de Muniz *et al.* 2010; Bravo 2010; Gimenez *et al.* 2015). Moglia & Gimenez (1998) found similar values to ours in only 7 of the 46 species they analyzed, all of which were shrubs and smaller trees. Furthermore, a comparison with the findings of Lindorf (1994) for species from arid regions of the world, shows similar results in species from Southern California flora and heath shrubs of Western Australia, as well as those of the extremely dry tropical forest of Mamo.

Among our species, vessel length was very variable. Most of them showed moderately-short vessels. Short vessels, which generally present thick cell walls, are associated with increased strength and resistance to high pressure and deformations resulting from water column tension (Lindorf 1994; Moglia & Gimenez 1998; Carlquist 2009). *F. thurifera* showed extremely short vessels, which might be compensating for its larger vessel diameter; conversely, *K. lanceolata* had vessels that were more than double the size of *F. thurifera*, possibly compensating for their shorter diameter.

The length, diameter, and frequency of the vessels determine the vulnerability and mesomorphy indexes. These indexes quantitatively assess the vulnerability and resistance of the wood and are used to classify species as either xero- or mesomorphic (Carquist 1977; Alemán-Sancheschúlz *et al.* 2019). All the species we analyzed had low degrees of vulnerability and high degrees of xeromorphism, with the lowest index values corresponding to *C. spinosissima* and *S. fasciculatus*. The five variables we analyzed had similar values to those reported by Carlquist (1977) for desert shrubs in North America, Arctic shrubs in Greenland, and sand heaths and desert shrubs in Western Australian Flora.

The grouping of vessels is vital in xeric environments as it increases hydraulic safety and prevents embolism, as a blocked vessel can be replaced by an adjacent one to maintain a continuous water flow (Lindorf 1994; Hargrave *et al.* 1995; Moglia & Gimenez 1998; Baas 2004; Carlquist 2009). Two species, *B. aliena* and *S. fasciculatus* presented a diagonal or dendritic vessel arrangement that interconnects vessels to form a wide network that improves water transport (Lindorf 1994; Baas 2004; Carlquist 2009). In *K. lanceolata* most of the vessels were solitary, but water efficiency could be aided by the high number of vasicentric tracheids observed in this species. According to Carlquist (2009), these are imperforate tracheids with bordered pits located around the vessels that increase interconnectivity and continuity of transport, and function as a subsidiary conductive system (Zhang 1992; Lindorf 1994; Hargrave *et al.* 1995). The presence of tracheids as a background tissue could be more effective than vessel grouping to prevent disabled conductive pathways (Hargrave *et al.* 1995; Carlquist 2009).

Growth rings were visible in all species and reflect the discontinuous activity of the cambium due to the climatic seasonality of the study area (Fritts 1976). *F. thurifera* was the only species that presented ring porosity, which allows rapid transport through the large vessels of the earlywood, while transport in the latewood is aided by small-diameter vessels. Ring porosity has been generally linked to arid environments (Carlquist 2009; Fahn & Cutler 1992). The other species presented diffuse porosity, which enables the whole ring to be available for water transport throughout the year. This characteristic is associated with mesic environments where seasonality is not clearly marked (Fritts 1976; Woodcock 1994).

The anatomical complexity of the bark reflects its many vital functions (Rosell 2019). A few layers of phellem and a single layer of phelloderm were present in the outer bark of all our species. These few layers of outer bark could be mainly involved in the protection of stems from diverse external hazards, such as interruption of photosynthate translocation, vessel deformation, and embolism due to high temperatures resulting from fires (Rosell *et al.* 2014; Rosell 2016, 2019), and also provide mechanical stem support (Pausas 2015; Rosell 2019). Bark thickness scales strongly with stem size (Rosell 2019), which explains the presence of a thin outer bark in the young branches of the stems we analysed (3–4 cm total diameter).

The inner living bark positively correlates with traits reflecting metabolic demand, such as leaf area, photosynthate translocation produced by secondary phloem, sapwood storage, and hydraulic conductivity (Rosell & Olson 2007; Rosell 2019). We found a developed inner bark in all the species, and in *C. spinosissima* also abundant photosynthetic chlorenchyma, which possibly compensates for the limited photosynthetic function produced by its deciduous leaves. We also found abundant fibre bundles around the secondary phloem of the inner bark in all species. These were particularly abundant in *B. aliena*, where the conducting elements of the secondary phloem were arranged in small groups surrounded by abundant libriform fibres forming a protective sheath. The presence of thick-walled cells in the inner bark gives protection and mechanical support to the stem that, given its peripheral location where bending and twisting stress are maximum, can substantially contribute to twig mechanics (Rosell 2019).

The presence of secretory structures in the wood and bark of *F. thurifera*, *S. fasciculatus*, and *K. lanceolata*, correlates with that reported for the leaves. Ducts, crystals, druses, and tannins are also common inner bark traits likely associated with defence against herbivores and pathogens (Roth 1981; Fahn 2002; Rosell 2019). Furthermore, resins and tyloses can obliterate damaged conductive cells or non-functional old rings to prevent water loss (Fahn & Cutler 1992; Lindorf 1994).

Functional coordination and trade-offs?

An integrative analysis including our results from this and our previous study shows that, except in the case of *Colletia spinosissima*, no species presented exclusively xeromorphic traits. Functional coordination and trade-offs between the leaf and the stem could explain the presence of a trait that in isolation could be considered detrimental in dry conditions, but when taken as a component of an interconnected and complementary system, would not necessarily and/or negatively affect survival (Mencuccini *et al.* 2019; Rosell 2019). Trade-offs, such as hydraulic safety vs efficiency, or quality vs quantity of seeds, are in fact ubiquitous (Stearns 1989) and occur because plants, like any other living organism, cannot optimize all functions simultaneously (Díaz *et al.* 2016; Walker *et al.* 2017).

Colletia spinosissima, *Baccharis aliena*, and *Schinus fasciculatus* were the species with the highest number of xeromorphic wood traits. They showed the highest number and smaller size of vessels per sq. mm, lowest IV, and highest MV. Leaf traits of these species were also xeromorphic. In *B. aliena* and *S. fasciculatus* leaves had a reduced surface area and a thick palisade parenchyma. In *C. spinosissima* the small leaves have mesomorphic characteristics, but they are rapidly deciduous, and assimilation is restricted to the thorny branches. Collectively, and despite the presence of dissimilar and non-xeric traits, these trait combinations allow these species to survive in arid areas and could, in principle, allow them to tolerate an increase in aridity and drought resulting from climate change.

Flourensia thurifera and *Kageneckia lanceolata* showed less xeromorphic wood and leaf traits. Their large leaves, a mesic trait associated with higher evapotranspiration, were possibly compensated by their higher stomatal density that increases the rate of gaseous exchange in environments with low humidity and high light intensity (Fahn & Cutler 1992). In addition, they present the ability to partially shed off their leaves in winter, when environmental temperature and water availability are low (Delbón *et al.* 2007a, b; Marticorena 2008; Romero & Ramos 2009; Ospina *et al.* 2018). *F. thurifera* additionally presents leaf sheaths and well-developed extensions, glandular trichomes, secretory ducts (Delbón *et al.* 2007 a, b), and underground xylopodia that enable re-sprouting after fires (Delbón *et al.* 2017). These particular characteristics of *F. thurifera* can complement one another and result in an alternative strategy to survive in changing environments. On the other hand, *K. lanceolata* did not show any xeromorphic leaf trait. We thus expect this species to live in areas where dry conditions are moderate or close to a water source (Romero & Ramos 2009; Romero 2019) and predict that it would not withstand a significant increase in aridity. The least-drought tolerant species are lost first when the climate becomes drier (Chapin & Díaz Bonham 2020) and their loss can affect the entire plant community as functional and phylogenetic diversity is removed (Harrison *et al.* 2020). Therefore, conservation efforts and strategies, such as conditioning of native species to drought-tolerance (Valliere *et al.* 2019), should be directed to species like *F. thurifera* and *K. lanceolata*.

CONCLUSION

Our findings provide further evidence of the diverse strategies that plants have developed to survive in dry conditions. Although most of the traits we describe here have already been reported in numerous studies, we highlight the importance of analysing how plants survive in different environments using an integrative approach. From this viewpoint, we can better understand and value the vast biodiversity of plants, and assess which species are more tolerant to environmental changes, but also which ones might be more vulnerable and thus require greater conservation efforts. Further research is required to fully understand how plant hydraulic traits are coordinated and interact from morpho-anatomical, physiological, and genetic perspectives. Given that plants are the main energy and matter entry point into the global ecosystem and that drought levels are expected to continue increasing, these remain critical areas of research.

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REFERENCES

Alemán-Sancheschúlz G, Solano E, Terrazas T, López-Portillo J. 2019. La arquitectura hidráulica de las plantas vasculares terrestres, una revisión. Madera Bosques 25(3): e253182825. DOI: 10.21829/myb.2019.2531828.

Anfodillo T, Petit G, Crivellaro A. 2013. Axial conduit widening in woody species: a still neglected anatomical pattern. IAWA J. 34: 352–364. DOI: 10.1163/9789004265608_004.

Ángeles G, Bond B, Boyer JS, Brodribb T, Brooks JR, et al. 2004. The cohesion-tension theory. New Phytol. 163: 451–452. DOI: 10.1111/j.1469-8137.2004.01142.x.

Angyalossy V, Pace MR, Evert RF, Marcati CR, Oskolski AA, Terrazas T, Kotina E, Lens F, Mazzoni-Viveiros SC, Angeles G, Machado SR, Crivellaro A, Rao KS, Junikka L, Nikolaeva N, Baas P. 2016. IAWA list of microscopic bark features. IAWA J. 37: 517–615.

Baas P. 2004. Evolution of xylem physiology. In: Hemsley AR, Poole I (eds.), The evolution of plant physiology: 273–291. Elsevier Academic Press, San Diego, CA.

Barboza GE, Cantero JJ, Nuñez CO, Ariza Espinar L. 2006. Flora medicinal de la provincia de Córdoba (Argentina). Ed. Museo Botánico Córdoba, Córdoba.

Blanco-Dios JB. 2007. Estudio morfométrico de una zona híbrida entre *Armeria beirana* y *A. pubigera* (Plumbaginaceae) en el noroeste de la Península Ibérica. Anales Jard. Bot. Madrid 64(2): 229–235. DOI: 10.3989/ajbm.2007.v64.i2.180.

Bolzon de Muniz GI, Nisgosky S, Lomelí-Ramírez MG. 2010. Anatomía y ultraestructura de la madera de tres especies de *Prosopis* (Leguminosae-Mimosoideae) del Parque Chaqueño seco, Argentina. Madera Bosques 16: 21–38.

Boyer JS. 1976. Water deficits and photosynthesis. In: Kozlowski TT (ed.), Water deficits and plant growth. Vol IV. Soil water measurement, plant responses, and breeding for drought resistance: 153–190. Academic Press, London.

Bradford KJ, Hsiao TC. 1982. Physiological responses to moderate water stress. In: Lange OL, Nobel PS, Osmond CB, Ziegler H (eds.), Physiological plant ecology II. Encyclopedia of plant physiology (new series). Vol. 12B. Springer, Berlin.

Bravo S. 2010. Anatomical changes induced by fire-damaged cambium in two native tree species of the Chaco region, Argentina. IAWA J. 31: 283–292. DOI: 10.1163/22941932-90000023.

Brodersen CR, McElrone AJ. 2013. Maintenance of xylem network transport capacity: a review of embolism repair in vascular plants. Front. Plant. Sci. 108: 1–11. DOI: 10.3389/fpls.2013.00108.

Brodersen CR, Roddy AB, Wason JW, McElrone AJ. 2019. Functional status of xylem through time. Annu. Rev. Plant Biol. 70: 407–433. DOI: 10.1146/annurev-arplant-050718-100455.

Brown A, Martinez Ortiz U, Acerbi M, Corcuera J. 2006. La Situación Ambiental Argentina 2005. Fundación Vida Silvestre, Buenos Aires.

Bucci SJ, Goldstein G, Meinzer FC, Scholz FG, Franco AC, Bustamante M. 2004. Functional convergence in hydraulic architecture and water relations of tropical savanna trees: from leaf to whole plant. Tree Physiol. 24: 891–899. DOI: 10.1093/treephys/24.8.891.

Cabido M, Zeballos SR, Zak M, Carranza ML, Giorgis MA, Cantero JJ, Acosta AT. 2018. Native woody vegetation in central Argentina: classification of Chaco and Espinal forests. Appl. Veg. Sci. 21: 298–311. DOI: 10.1111/avsc.12369.

Carlquist S. 1977. Ecological factors in wood evolution: a floristic approach. Am. J. Bot. 64: 887–896. DOI: 10.1002/j.1537-2197.1977.tb11932.x.

Carlquist S. 2009. Non-random vessel distribution in woods: patterns, modes, diversity, correlations. Aliso 27: 39–58. DOI: 10.5642/aliso.20092701.04.

Castello L, Galetto L. 2013. How many taxa can be recognized within the complex *Tillandsia capillaris* (Bromeliaceae, Tillandsioideae)? Analysis of the available classifications using a multivariate approach. PhytoKeys 23: 25–39. DOI: 10.3897/phytokeys.23.4507.

Chapin FS III, Díaz S. 2020. Interactions between changing climate and biodiversity: shaping humanity's future. Proc. Natl. Acad. Sci. USA 117: 6295–6296. DOI: 10.1073/pnas.2001686117.
Deble LP, de Oliveira AS, Marchiori JNC. 2005. O gênero *Heterothalamus* Lessing e táxones afins. Balduinia 1: 01–20. DOI: 10.5902/2358198014003.
Delbón N, Cortez MA, Castello L, Rios A, Risso MJ, Dottori N, Cosa MT. 2010. Anatomía foliar y estrategias adaptativas en especies arbustivas de las Sierras Chicas de Córdoba, Argentina. Arnaldoa 17: 41–49.
Delbón N, Cosa MT, Dottori N. 2007a. Anatomía de órganos vegetativos en *Flourensia campestris* y *F. oolepis* (Asteraceae), con especial referencia a las estructuras secretoras. Arnaldoa 14: 61–70.
Delbón N, Cosa MT, Dottori N, Stiefkens L. 2007b. Estudio de la epidermis foliar en *Flourensia campestris* y *F. oolepis* (Asteraceae). Bol. Soc. Argent. Bot. 42: 245–250.
Delbón N, Cosa MT, Bernardello G. 2017. Reproductive biology, seed germination and regeneration of *Flourensia* DC. species endemic to Central Argentina (Asteraceae). Adansonia 39: 71–85. DOI: 10.5252/a2017n1a6.
Denham S, Múlgura de Romero ME, Slanis A, Bulacio E. 2006. *Lippia integrifolia* versus *L. boliviana* (Verbenaceae). Darwiniana 44: 363–374.
Díaz S, Kattge J, Cornelissen JHC, Wright IJ, Lavorel S, et al. 2016. The global spectrum of plant form and function. Nature 529: 167–171. DOI: 10.1038/nature16489.
Díaz S, Settele J, Brondizio ES, Ngo HT, Agard J, et al. 2019. Pervasive human-driven decline of life on Earth points to the need for transformative change. Science 366: eaax3100. DOI: 10.1126/science.aax3100.
Diffenbaugh NS, Swain DL, Touma D. 2015. Anthropogenic warming has increased drought risk in California. Proc. Natl. Acad. Sci. USA 112: 3931–3936. DOI: 10.1073/pnas.1422385112.
Fahn A, Cutler DF. 1992. Xerophytes. Handbuchder Pflanzenanatomie XIII, 3. Gebrüder Borntraeger, Berlin.
Fahn A. 2002. Functions and location of secretory tissues in plants and their possible evolutionary trends. Isr. J. Plant Sci. 50: 59–64. DOI: 10.1560/LJUT-M857-TCB6-3FX5.
Flach M, Brenning A, Gans F, Reichstein M, Sippel S, Mahecha MD. 2020. Vegetation modulates the impact of climate extremes on gross primary production. Biogeosciences. Preprint. DOI: 10.5194/bg-2020-80.
Fritts HC. 1976. Tree rings and climate. Academic Press, New York, NY.
Fu X, Meinzer FC, Woodruff DR, Liu Y-Y, Smith DD, McCulloh KA, Howard AR. 2019. Coordination and trade-offs between leaf and stem hydraulic traits and stomatal regulation along a spectrum of isohydry to anisohydry. Plant Cell Environ. 42: 2245–2258. DOI: 10.1111/pce.13543.
Gimenez AM, Calatayu F, Zirpolo JD, Figueroa ME, Gonzalez D. 2015. Anatomía comparada del leño de tres especies nativas de *Schinopsis* (Anacardiaceae). Bol. Soc. Argent. Bot. 50: 323–335. DOI: 10.31055/1851.2372.v50.n3.12522.
Giorgis MA, Cingolani AM, Chiarini F, Chiapella J, Barboza G, Ariza Espinar L, Morero R, Gurvich D, Tecco PA, Subils R, Cabido MR. 2011. Composición florística del Bosque Chaqueño Serrano de la Provincia de Córdoba, Argentina. Kurtziana 36: 9–43.
Global Forest Watch. 2020. 'Tree cover loss in Argentina'. Available online at www.globalforestwatch.org (accessed 1 April 2020).
Hargrave KR, Kolb KJ, Ewers FW, Davis SD. 1995. Conduit diameter and drought-induced embolism in *Salvia mellifera* Greene. New Phytol. 126: 695–705. DOI: 10.1111/j.1469-8137.1994.tb02964.x.
Harrison S, Spasojevic MJ, Li D. 2020. Climate and plant community diversity in space and time. Proc. Natl. Acad. Sci. USA 117: 4464–4470. DOI: 10.1073/pnas.1921724117.
IAWA Committee. 1989. IAWA list of microscopic features for hardwood identification with an appendix on non-anatomical information. IAWA Bull. n.s. 10: 219–332. DOI: 10.1163/22941932-90000496.
Jiao T, Williams CA, Rogan J, DeKauwe MG, Medlyn BE. 2020. Drought impacts on Australian vegetation during the millennium drought measured with multisource spaceborne remote sensing. J. Geophys. Res-Biogeogr. 125: e2019JG005145. DOI: 10.1029/2019JG005145.
Kawecki TJ, Ebert D. 2004. Conceptual issues in local adaptation. Ecol. Lett. 7: 1225–1241. DOI: 10.1111/j.1461-0248.2004.00684.x.
Kraus JE, de Sousa HC, Rezende MH, Castro NM, Vecchi C, Luque R. 1998. Astra blue and basic fuchsin double staining of plant materials. Biotechnol. Histochem. 73: 235–243. DOI: 10.3109/10520299809141117.
Lachenbruch B, McCulloh KA. 2014. Traits, properties, and performance: how woody plants combine hydraulic and mechanical functions in a cell, tissue, or whole plant. New Phytol. 204: 747–764. DOI: 10.1111/nph.13035.
Lindorf H. 1994. Eco-anatomical wood features of species from a very dry tropical forest. IAWA J. 15: 361–376. DOI: 10.1163/22941932-90001370.
Loram-Lourenço L, Farnese FdS, Sousa LFd, Alves RDFB, Andrade MCPd, Almeida SEdS, Moura LMdF, Costa AC, Silva FG, Galmés J, Cochard H, Franco AC, Menezes-Silva PE. 2020. A structure shaped by fire, but also water: ecological consequences of the variability in bark properties across 31 species from the Brazilian Cerrado. Front. Plant Sci. 10: 1718. DOI: 10.3389/fpls.2019.01718.
MacLachlan IR, Gasson P. 2010. PCA of CITES listed *Pterocarpus santalinus* (Leguminosae) wood. IAWA J. 31(2): 121–138. DOI: 10.1163/22941932-90000010.
Marticorena A. 2008. In: Zuloaga FO, Morrone MO, Belgrano MJ (eds.), Catálogo de las Plantas Vasculares del Cono Sur. Vol. 3. Missouri Botanical Garden, St. Louis, MO.
Martijena N. 1987. Wood anatomy of *Lithraea ternifolia* (Gill.) Barkley' Rom. (Anacardiaceae). IAWA J. 8: 47–52. DOI: 10.1163/22941932-90001025.
Martín-Sanz RC, San-Martín R, Poorter H, Vázquez A, Climent J. 2019. How does water availability affect the allocation to bark in a Mediterranean conifer? Front. Plant Sci. 10: 607. DOI: 10.3389/fpls.2019.00607.
Mencuccini M, Rosas T, Rowland L, Choat B, Cornelissen H, et al. 2019. Leaf economics and plant hydraulics drive leaf: wood area ratios. New Phytol. 224: 1544–1556. DOI: 10.1111/nph.15998.

Miranda JD, Padilla FM, Martinez-Vilalta J, Pugnaire FI. 2010. Woody species of a semi-arid community are only moderately resistant to cavitation. Funct. Plant Biol. 37: 828–839. DOI: 10.1071/FP09296.

Moglia G, Jiménez AM. 1998. Rasgos anatómicos característicos del hidrosistema de las principales especies arbóreas de la región Chaqueña Argentina. Invest. Agrar. Sist. Rep. 7: 53–71.

Neyses B, Hagman O. 2014. Identifying suitable wood species for wooden products with multivariate data analysis. In: Proceedings of the 4th international conference of production engineering and management, September: 25–26.

Olson ME, Soriano D, Rosell JA, Anfodillo T, Donoghue MJ, Edwards EJ, Echeverría A. 2018. Plant height and hydraulic vulnerability to drought and cold. Proc. Natl. Acad. Sci. USA 115: 7551–7556. DOI: 10.1073/pnas.1721728115.

Oskolski AA, Lowry II PP. 2001. Wood anatomy of *Schefflera* and related taxa (Araliaceae). II. Systematic wood anatomy of New Caledonian *Schefflera*. IAWA J. 22: 301–330.

Ospina JC, Aagesen L, Ariza Espinar L, Freire SE. 2018. Morphometric analyses and new taxonomic circumscription of South American species of *Flourensia* (Asteraceae, Heliantheae, Enceliinae). Nord. J. Bot. 36: 1–15. DOI: 10.1111/njb.01737.

Pausas JG. 2015. Bark thickness and fire regime. Funct. Ecol. 29: 315–327. DOI: 10.1111/1365-2435.12372.

Pittermann J. 2010. The evolution of water transport in plants: an integrated approach. Geobiology 8: 112–139. DOI: 10.1111/j.1472-4669.2010.00232.x.

Primack R, Rozzi R, Feisinger P, Dirzo R, Massardo F. 2001. Fundamentos de conservación biológica, 1st Edn. Fondo de cultura económica, México City.

Romero C, Ramos D. 2009. Composición florística y estado de conservación de los bosques de *Kageneckia lanceolata* Ruiz & Pav. y *Escallonia myrtilloides* L.F. en la reserva paisajística Nor Yauyos Cochas. Tesis para optar el título profesional de ingeniero forestal y ambiental, Universidad Nacional del Centro del Perú, Facultad de Ciencias Forestales y del Ambiente, Huancayo-Junín.

Romero M. 2019. Distribución y asociación del "lloque" *Kageneckia lanceolata* Ruiz & Pavón en la cuenca del río Canchacalla, San Mateo de Otao (Huarochiri, Lima). Tesis para optar al título profesional de ingeniero agroforestal, Universidad Científica del Sur, Facultad de Ciencias Ambientales, Lima.

Rosell JA, Olson ME. 2007. Testing implicit assumptions regarding the age vs. size dependence of stem biomechanics using *Pittocaulon* (*Senecio*) *praecox* (Asteraceae). Am. J. Bot. 94: 161–172. DOI: 10.3732/ajb.94.2.161.

Rosell JA, Gleason S, Méndez-Alonzo R, Chang Y, Westoby M. 2014. Bark functional ecology: evidence for tradeoffs, functional coordination, and environment producing bark density. New Phytol. 201: 484–497. DOI: 10.1111/nph.12541.

Rosell JA. 2016. Bark thickness across the angiosperms: more than just fire. New Phytol. 211: 90–102. DOI: 10.1111/nph.13889.

Rosell JA. 2019. Bark in woody plants: understanding the diversity of a multifunctional structure. Integ. Comp. Biol. 59: 535–547. DOI: 10.1093/icb/icz057.

Roth I. 1981. Structural patterns of tropical barks. Encyclopedia of plant anatomy. Gebrüder Borntraeger, Berlin.

Scheffé H. 1959. The analysis of variance. Wiley, New York.

Schweingruber FH, Börner A. 2018. The plant stem: a microscopic aspect. Springer, Cham.

Sobrado MA. 1993. Trade-off between water transport efficiency and leaf life-span in a tropical dry forest. Oecologia 96: 19–23. DOI: 10.1007/BF00318025.

Sokal RR, Rohlf FJ. 1981. Biometry: the principles and practice of statistics in biological research. Freeman, San Francisco, CA.

Sperry JS, Tyree MT. 1988. Mechanism of water stress-induced xylem embolism. Plant Phys. 88: 581–587.

Sperry JS. 2003. Evolution of water transport and xylem structure. Int. J. Plant Sci. 164: 115–127. DOI: 10.1086/368398.

Stearns SC. 1989. Trade-offs in life-history evolution. Funct. Ecol. 3: 259–268.

Steudle E. 2001. The cohesion-tension mechanism and the acquisition of water by plant roots. Annu. Rev. Plant Biol. 52: 847–875.

Tortosa RD. 1989. El Género *Colletia* (Rhamnaceae). Parodiana 5: 279–332.

Tortosa RO. 2008. In: Zuloaga FO, Morrone MO , Belgrano MJ (eds.), Catálogo de las Plantas Vasculares del Cono Sur. Vol. 2. Missouri Botanical Garden, St. Louis, MO.

Valliere JM, Zhang J, Sharifi MR, Rundel PW. 2019. Can we condition native plants to increase drought tolerance and improve restoration success? Ecol. Appl. 29: e01863. DOI: 10.1002/eap.1863.

Valtierra V, Bonifacino JM. 2014. Revisión taxonómica de *Baccharis* sect. *Heterothalamus* (Asteraceae: Astereae) en Uruguay. Bol. Soc. Argent. Bot. 49: 613–620. DOI: 10.31055/1851.2372.v49.n4.9993.

Vicente-Serrano SM, López-Moreno JI, Beguería S, Lorenzo-Lacruz J, Sanchez-Lorenzo A, Garcia-Ruiz JM, Azorin-Molina C, Morán-Tejeda E, Revuelto J, Trigo R, Coelho F, Espejo F. 2014. Evidence of increasing drought severity caused by temperature rise in southern Europe. Environ. Res. Lett. 9: 044001. DOI: 10.1088/1748-9326/9/4/044001.

Villagra PE, Roig Juñent FA. 1997. Wood structure of *Prosopis alpataco* and *P. argentina* growing under different edaphic conditions. IAWA J. 18: 37–51. DOI: 10.1163/22941932-90001458.

Walker AP, McCormack ML, Messier J, Myers-Smith IH, Wullschleger SD. 2017. Trait covariance: the functional warp of plant diversity? New Phytol. 216: 976–980. DOI: 10.1111/nph.14853.

Whitlock MC. 2015. Modern approaches to local adaptation. Am. Nat. 186: 1–4.

Woodcock DW. 1994. Occurrence of woods with a gradation in vessel diameter across the ring. IAWA J. 15: 377–385. DOI: 10.1163/22941932-90001371.

Woods K, Hilu KW, Wiersema JH, Borsch T. 2005. Pattern of variation and systematics of *Nymphaea odorata*: I. Evidence from morphology and inter-simple sequence repeats (ISSRs). Syst. Bot. 30: 471–480. DOI: 10.1600/0363644054782161.

Zarlavsky GE. 2014. Histología vegetal. Técnicas simples y complejas. Soc. Argent. Bot., Buenos Aires.

Zhang F, Quan Q, Ma F, Tian D, Hoover DL, Zhou Q, Niu S. 2019. When does extreme drought elicit extreme ecological responses? J. Ecol. 107: 2553–2563. DOI: 10.1111/1365-2745.13226.
Zhang S. 1992. Systematic wood anatomy of the Rosaceae. Blumea 37: 81–158.
Zuloaga FO, Morrone MO, Belgrano MJ. 2008. Catálogo de las Plantas Vasculares del Cono Sur. Vol. 2. Missouri Botanical Garden, St. Louis, MO.

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APPENDIX

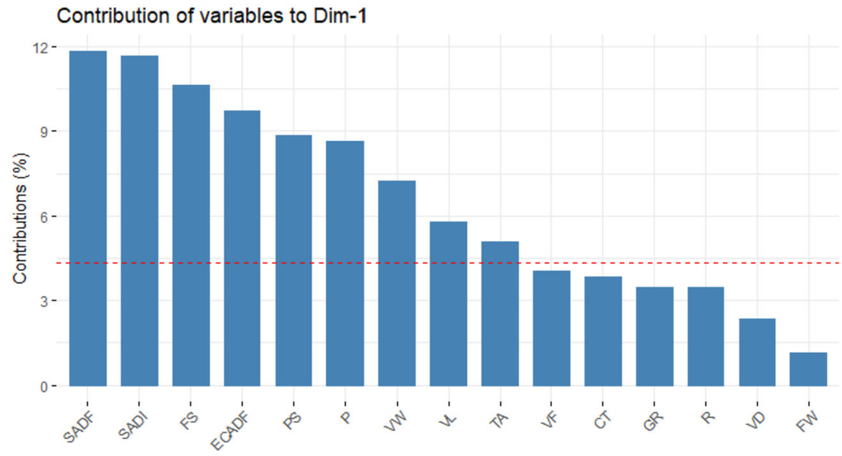

Figure A1. The expected contribution to the first PC from the PCA analyses of all the wood and leaf traits. The variables that were above the cut-off line were included in a second PCA analysis with a reduced matrix. Abbreviations of variables: CT, cuticle thickness; ECADF, adaxial epidermal cells frequency; FS, foliar structure; FW, fibre width; GR, growth ring boundaries; P, porosity; PS, palisade/spongy ratio; R, rays; SADF, adaxial stomatal frequency; SADI, adaxial stomatal index; TA, trichome abundance; VD, vessel disposition; VF, vessels per sq. mm; VL, vessel length; VW, vessel width.

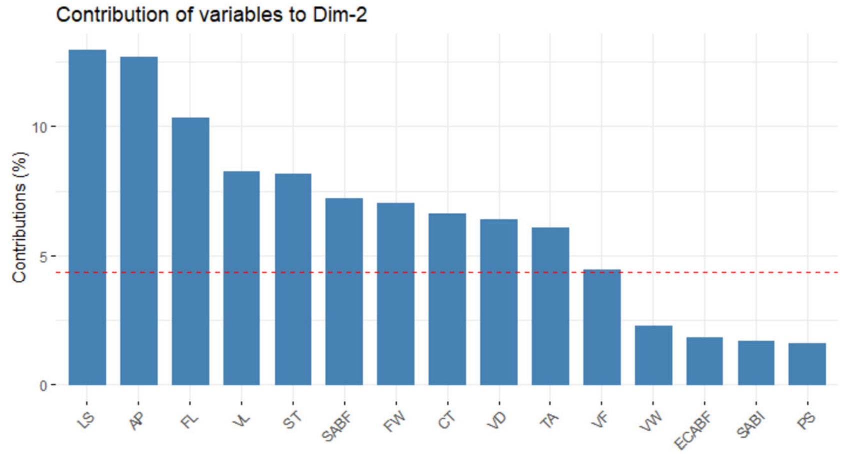

Figure A2. The expected contribution to the second PC from the PCA analyses of all the wood and leaf traits. The variables that were above the cut-off line were included in a second PCA analysis with a reduced matrix. Abbreviations of variables: AP, axial parenchyma; CT, cuticle thickness; ECABF, abaxial epidermal cells frequency, SABI, abaxial stomatal index; FL, fibre length; FW, fibre width; LS, leaf size; PS, palisade/spongy ratio; SABF, abaxial stomatal frequency; ST, stomatal type; TA, trichome abundance; VD, vessel disposition; VF, vessels per sq. mm; VL, vessel length; VW, vessel width.